

Enhancing Time Series Forecasting through Multi-Modal Data Fusion

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Motivation: Problem setting

Time-series forecasting underlies critical decisions in finance, public health, and infrastructure.

Traditional Time-series forecasting relies solely on historical numerical data.

Motivation: Why Text?

- **Textual data encodes information about external events.**
- Reports, news, and documents provide semantic context behind numerical changes.
- Text can be viewed as **side information** revealing latent drivers.
- Incorporating text expands the **observable information space**.

Motivation: Key Idea

- Standard forecasting: $y_{t+1} = f(x_{1:t})$
- Event-aware forecasting: $y_{t+1} = f(x_{1:t}, \text{context}_t)$
- **Goal: reduce uncertainty by integrating semantic context.**
- This motivates a **multimodal forecasting framework.**

Problem Setup: Multimodal Forecasting

Inputs

- Numerical time series: $\mathcal{X} = \{x_1, \dots, x_T\}$, $x_t \in \mathbb{R}^C$
- Aligned text context: $\mathcal{S} = \{s_1, \dots, s_T\}$

Sliding-window formulation

$$\mathbf{X}_{in} = \{x_{t-L+1}, \dots, x_t\}, \quad \mathbf{S}_{in} = \{s_{t-L+1}, \dots, s_t\}$$

$$\hat{\mathbf{Y}} = \mathcal{F}(\mathbf{X}_{in}, \mathbf{S}_{in}; \Theta), \quad \mathbf{Y} = \{x_{t+1}, \dots, x_{t+H}\}$$

Goal: leverage semantic signals to reduce error under *event-driven volatility* (regime shifts, shocks).

Baseline: MM-TSFlib Late Fusion (Time-MMD)

Two-branch pipeline

- Trainable TS backbone: $\text{TSF}(X)$
- Frozen text encoder (BERT): $E_{\text{text}} = \text{BERT}(S) + \text{trainable projection}$

Late fusion at output

$$\hat{Y} = (1 - \lambda) \cdot \text{TSF}(X) + \lambda \cdot (\text{Proj}(E_{\text{text}}) + Y_{\text{prior}})$$

Limitations (why we change it)

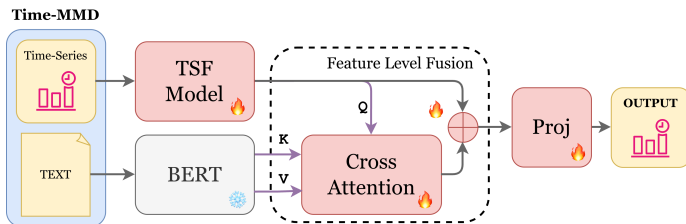
- **Shallow fusion**: modalities don't interact inside the feature space
- **Global static weight** λ : assumes constant modality importance
- **Prior dependence** Y_{prior} : text branch behaves like an offset

Our Idea: Feature-Level Fusion via Cross-Attention + Gating

Key change: fuse *features* before prediction head (not just outputs).

- Temporal encoder produces hidden states: $\mathbf{H}_{ts} \in \mathbb{R}^{L \times D}$
- Frozen text encoder produces embeddings: $\mathbf{H}_{text} \in \mathbb{R}^{S \times D_{text}}$
- Cross-Attention injects semantic context into temporal features
- Adaptive gating learns *time-step specific* fusion weights

Time cross attention architecture:



Cross-Modal Attention: Q from TS, K/V from Text

Projections (dimension alignment)

$$Q = \text{LN}(\mathbf{H}_{ts})\mathbf{W}_Q, \quad K = \text{LN}(\mathbf{H}_{text})\mathbf{W}_K, \quad V = \text{LN}(\mathbf{H}_{text})\mathbf{W}_V$$

$$\mathbf{W}_Q \in \mathbb{R}^{D \times D}, \quad \mathbf{W}_K, \mathbf{W}_V \in \mathbb{R}^{D_{text} \times D}$$

Multi-Head Cross-Attention

$$\tilde{\mathbf{H}} = \text{MHA}(Q, K, V) = \text{Softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right) V$$

Residual + normalization + output projection

$$\mathbf{H}_{attn} = \text{Proj}_{out}\left(\text{LN}(Q + \tilde{\mathbf{H}})\right)$$

Intuition: numerical states query the most relevant semantic cues (event descriptions) that explain changes.

Adaptive Temporal Gating: Dynamic Fusion (No Global λ)

Learned gate per time step

$$\alpha = \sigma(\mathbf{G}), \quad \mathbf{G} \in \mathbb{R}^{L \times 1}$$

Fuse temporal vs semantic-injected features

$$\mathbf{Z} = (1 - \alpha) \odot \mathbf{H}_{ts} + \alpha \odot \mathbf{H}_{attn}$$

Forecast head

$$\hat{Y} = \text{Proj}_{final}(\mathbf{Z})$$

- $\alpha \rightarrow 0$: trust historical numeric patterns
- $\alpha \rightarrow 1$: emphasize semantic context (shock / event periods)

Training & Practical Details (Implementation-Ready)

Optimization objective

$$\mathcal{L} = \frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2 \quad (\text{MSE}), \quad \text{also report MAE}$$

Implementation choices

- Text encoder frozen (BERT); only projection + fusion layers trained
- Text window: tail segment of lookback (e.g., last 4 steps) + attention pooling
- Chronological split: Train/Val/Test = 70/10/20 (avoid leakage)
- Standardization computed on train set only: $\tilde{x}_t = \frac{x_t - \mu_{train}}{\sigma_{train}}$

Why this is stable: feature-level fusion + gating reduces “semantic noise” injection in non-event regions.

Why Informer Helps but Reformer Can Break

Observed phenomenon

- Informer (ProbSparse) + semantic injection: often improves
- Reformer (LSH) + semantic injection: can degrade in periodic domains (e.g., Traffic)

One-line mechanism

- Informer selects “active queries” globally; semantic features make event steps more distinctive
- Reformer relies on locality in embedding space; semantic displacement can break LSH bucketing

Methodology Takeaways

- **Baseline issue:** late fusion cannot learn deep cross-modal representations
- **Our fix:** cross-attention injects text into TS features *before* prediction
- **Robustness:** adaptive gate learns when text is helpful vs harmful
- **Compatibility insight:** fusion interacts with backbone attention type (global vs LSH locality)

Transition: next, we evaluate on Time-MMD across domains/horizons and compare against unimodal + MM-TSFlib.

Innovation 3: Backbone-Aware Multimodal Insight

- Multimodal fusion does not interact equally with all attention mechanisms.
- We observe strong synergy with **Informer (global attention)**.
- We identify negative transfer with **Reformer (LSH-based attention)**.
- We explain this via **Semantic Manifold Distortion**.

Finding 2: Architecture Compatibility Matters

Informer: Synergy

Traffic Dataset (MSE)

Len	Uni	Ours
3	0.246	0.194
6	0.257	0.197
9	0.304	0.194

✓ Consistent gains

Reformer: Conflict

Traffic Dataset (MSE)

Len	Uni	Ours
3	0.203	0.256
6	0.233	0.253
9	0.202	0.322

× Degradation

Root Cause: Semantic injection enhances ProbSparse attention but distorts LSH manifold in periodic domains

Finding 3: The Mid-Horizon Sweet Spot

Economy Dataset (Informer)

Len	Uni		Ours	
	MSE	MAE	MSE	MAE
3	0.558	0.697	0.652	0.775
6	1.043	0.978	0.861	0.895
9	1.158	1.035	0.904	0.923
12	0.777	0.848	1.298	1.113

Pattern

- L=3: Numerical sufficient
- L=6,9: Semantic wins
- L=12: Semantic decays

Textual signals act as *leading indicators* with time-bounded validity

Conclusion & Future Work

Key Takeaways

- Cross-attention fusion enables semantic-temporal alignment
- **Not universal:** Depends on domain volatility, architecture, and horizon
- Design principle: Match fusion to backbone and periodicity

Future Directions

- Hybrid fusion strategies
- Multi-granular text alignment
- Domain-adaptive encoders

Thank You!

Code: <https://github.com/xlesley/ese-5380-11xx>